Day vs. Night Object detection: YOLOv8 on BDD100K

Rishab Choudhary

***Abstract*—Reliable object detection is essential for autonomous driving, but variations in lighting conditions can affect detection performance. This study investigates how different training strategies influence the accuracy of YOLOv8n on the BDD100K driving dataset. Four experiments are considered: a COCO-pretrained baseline (E0), balanced daytime and nighttime training (E1), nighttime-only training (E2), and daytime-only training (E3). The fine-tuned models are trained for 20 epochs and evaluated separately on daytime and nighttime validation images using mean Average Precision (mAP), precision, and recall. Results show that balanced training achieves the highest nighttime mAP50 of 47.91%, compared with 39.48% for nighttime-only training and 37.43% for daytime-only training. The balanced model also achieves 45.23% daytime mAP50, demonstrating relatively consistent performance across lighting conditions. In contrast, condition-specific models exhibit larger performance differences between daytime and nighttime environments. These findings suggest that incorporating diverse lighting conditions during training may improve detection performance across varying illumination conditions, although further evaluation is needed to confirm generalization.**

***Index Terms*—Object Detection, YOLOv8n, BDD100K, Daytime and Nighttime Detection ,Transfer Learning**

I. Introduction

AUTONOMOUS driving systems depend on reliable visual perception to understand their surroundings and make safe driving decisions. Object detection is an essential component of this process, enabling vehicles to identify and locate road users such as cars, pedestrians, cyclists, buses, and trucks. Although modern deep learning models have achieved significant progress in object detection, their performance can vary considerably depending on the environmental conditions in which they operate.

One important challenge is the variation in lighting between daytime and nighttime driving. During the day, objects are generally more visible due to sufficient natural illumination. At night, however, reduced visibility, uneven artificial lighting, headlight glare, and low contrast can make it difficult to distinguish objects from their surroundings. These changes can affect the visual features learned by object detection models and potentially reduce their reliability in real-world driving scenarios. Therefore, evaluating detection performance under different illumination conditions is important when developing robust perception systems.

This project investigates the influence of the composition of training-data on object detection performance in day and night environments. We employ YOLOv8n [2], a lightweight object detection model initialized with weights pretrained on the COCO dataset, and use the BDD100K driving dataset [1] for evaluation and fine-tuning. BDD100K provides driving images with time-of-day annotations, allowing daytime and nighttime scenarios to be studied separately.

To investigate the effect of different training strategies, four experimental configurations are considered: a pretrained baseline without BDD100K fine-tuning (E0), a model trained using a balanced combination of daytime and nighttime images (E1), a model trained exclusively on nighttime images (E2), and a model trained exclusively on daytime images (E3). The models are evaluated using standard object detection metrics, including mean Average Precision (mAP), precision, and recall, to examine both detection accuracy and differences in performance across lighting conditions.

The central research question of this study is: *How does the composition of daytime and nighttime training data influence the detection performance and cross-condition generalization of YOLOv8n on the BDD100K dataset?*

By comparing these training configurations, this study aims to determine whether exposure to both lighting conditions provides more consistent detection performance than training on a single lighting condition. The findings contribute to understanding the practical trade-offs involved in adapting lightweight object detectors to visually diverse driving environments.

II. Background Theory and Related Work

A. Object Detection

Object detection is a computer vision task that involves identifying objects in an image and determining their locations using bounding boxes. Unlike image classification, which assigns a label to an entire image, object detection predicts both the category and position of individual objects. In autonomous driving, this allows a perception system to recognize vehicles, pedestrians, and cyclists within the surrounding traffic environment.

Deep learning-based object detectors are commonly divided into two-stage and one-stage approaches. Two-stage detectors first identify candidate object regions and then classify them, while one-stage detectors perform localization and classification within a single detection network. The latter approach is particularly useful when processing speed is important, as in driving applications.

B. YOLOv8 Object Detection Model

The You Only Look Once (YOLO) family of models is widely used for object detection because it combines object localization and classification in an efficient architecture.

YOLOv8, developed by Ultralytics, introduces an anchor-free detection head and uses feature representations at multiple scales to detect objects of different sizes [2].

In this project, YOLOv8n, the nano version of YOLOv8, is selected because of its relatively small model size and lower computational requirements. This makes it suitable for experiments with limited GPU resources. The model is initialized using weights pretrained on the COCO dataset [3]. These weights provide a starting point for detecting common object categories before the model is fine-tuned on driving images.

C. BDD100K Dataset

BDD100K was introduced by Yu et al. [1] as a large-scale dataset for computer vision tasks related to autonomous driving. It contains driving scenes recorded under different weather, road, and lighting conditions. The dataset also provides object annotations and scene information, including time-of-day labels.

This variation makes BDD100K useful for studying how object detection models behave under different environmental conditions. For this project, five object categories are considered: person, bicycle, car, bus, and truck. The available time-of-day annotations are used to separate daytime and nighttime images, allowing the detection performance to be compared across the two conditions.

D. Object Detection Under Low-Light Conditions

Lighting conditions can significantly influence the visual information available to an object detector. In nighttime images, reduced contrast and uneven illumination may make object boundaries less distinguishable. Headlights, reflections, and poorly illuminated road users can introduce additional difficulties.

Loh and Chan [4] investigated object detection in low-light environments using the Exclusively Dark (ExDark) dataset. Their work showed that low illumination affects the visual features used for object recognition, highlighting the need to consider lighting conditions when evaluating detection models.

While previous research has examined the challenges of low-light object detection, this project focuses on the effect of training-data composition. Rather than introducing a new detection architecture or image enhancement method, we compare the same YOLOv8n model under different fine-tuning configurations. These include balanced day/night training, nighttime-only training, and daytime-only training, along with a pretrained baseline. Evaluating each configuration on separate daytime and nighttime image sets allows us to examine how training conditions influence detection accuracy.

III. Methodology

A. Dataset Preparation

The experiments use the BDD100K driving dataset [1], which contains annotated road scenes captured under different lighting and environmental conditions. Images are divided into daytime and nighttime groups using the dataset's time-of-day annotations. Five object categories are selected: person, bicycle, car, bus, and truck.

TABLE I
Evaluation Dataset Distribution

Subset	Day	Night	Total
Validation	750	750	1500
test	750	750	1500

TABLE II
Experimental Training Configurations

Exp.	Day	Night	Total	Epochs
E0	0	0	0	0
E1	3500	3500	7000	20
E2	0	7000	7000	20
E3	7000	0	7000	20

Three training configurations are considered. The balanced dataset contains 3,500 daytime and 3,500 nighttime images, while the nighttime-only and daytime-only datasets contain 7,000 images each. Training images are selected from the official BDD100K training split. Another 3,000 images are taken from the official validation split and divided equally into validation and test sets. Each evaluation set contains an equal number of daytime and nighttime images.

The original bounding-box annotations are converted into YOLO format, where each object is represented by a class ID and normalized bounding-box coordinates. The five selected categories are mapped to class IDs from 0 to 4. All images are processed at a resolution of 640×640 pixels.

B. Model Architecture and Training

YOLOv8n is selected as the object detection model because of its lightweight architecture and efficient detection process [2]. The network consists of a backbone that extracts visual features, a neck that combines information at different scales, and an anchor-free detection head that predicts object classes and bounding boxes.

Transfer learning is used by initializing the models with COCO-pretrained weights [3]. This allows the detector to build on previously learned visual features while adapting to the five selected BDD100K classes.

Four experimental configurations are compared. E0 represents the pretrained model without additional fine-tuning. E1 is trained on a balanced combination of daytime and nighttime images. E2 is trained exclusively on nighttime images, while E3 uses only daytime images.

The three fine-tuned models follow the same intended training configuration, consisting of 20 epochs, an image resolution of 640×640, and a random seed of 42.

C. Evaluation Procedure

All models are evaluated separately on the same 750 daytime and 750 nighttime validation images. Detection performance is measured using precision, recall, mAP@0.50, and mAP@0.50:0.95.

TABLE III
VALIDATION PERFORMANCE AND DAY–NIGHT GAP (%)

Model	mAP@0.50		Gap	mAP@0.50:0.95	
	Day	Night	(pp)	Day	Night
E0	33.53	24.48	+9.05	20.45	14.27
E1	**45.23**	**47.91**	-2.68	27.04	**28.34**
E2	32.15	39.48	-7.33	18.74	22.76
E3	44.74	37.43	+7.31	**27.14**	21.57

Precision measures the proportion of predicted detections that are correct, while recall measures the proportion of ground-truth objects successfully detected. Intersection over Union (IoU) measures the overlap between predicted and ground-truth bounding boxes. The mAP@0.50 metric averages Average Precision across the five object classes at an IoU threshold of 0.50, while mAP@0.50:0.95 averages results across thresholds from 0.50 to 0.95.

For the pretrained baseline E0, the selected BDD100K categories are mapped to their corresponding COCO class IDs. The fine-tuned models are evaluated using the five-class BDD100K configuration.

Comparing the daytime and nighttime results allows us to examine whether training on both lighting conditions provides more consistent detection performance than training on a single condition. The test set is kept separate and is not included in the reported validation results.

IV. RESULTS AND DISCUSSION

A. Overall Detection Performance

All four models are evaluated on the same validation images, consisting of 750 daytime and 750 nighttime scenes. Performance is measured using mAP@0.50 and mAP@0.50:0.95. Table III summarizes the results, including the difference between daytime and nighttime mAP@0.50.

The pretrained baseline E0 achieves 33.53% daytime and 24.48% nighttime mAP@0.50, showing a 9.05 percentage-point decrease at night. This indicates that the pretrained model performs less effectively on nighttime images in the selected validation data.

Among the fine-tuned models, E1 achieves the highest mAP@0.50 for both conditions, reaching 45.23% during daytime and 47.91% at night. Compared with E0, this corresponds to improvements of 11.70 and 23.43 percentage points, respectively.

B. Day–Night Performance Gap

To measure the difference between lighting conditions, the day–night gap is defined as

$$\Delta_{\mathrm{DN}} = \mathrm{mAP}_{\mathrm{Day}} - \mathrm{mAP}_{\mathrm{Night}}. \tag{1}$$

A positive gap indicates higher daytime accuracy, while a negative gap indicates higher nighttime accuracy. A smaller absolute gap indicates more similar performance across the two evaluation subsets.

E0 has a positive gap of 9.05 percentage points, confirming a noticeable drop in nighttime performance. After balanced fine-tuning, E1 has a gap of -2.68 points, reducing the absolute difference by 6.37 points compared with E0. This is a substantial reduction in the observed disparity between lighting conditions.

The single-condition models show larger gaps. E2 has a gap of -7.33 points, indicating better performance at night, while E3 has a gap of +7.31 points, indicating better performance during daytime. These opposite patterns are consistent with the lighting conditions used for their training.

An unexpected result is that E1 performs slightly better at night than during daytime. However, the smaller gap does not necessarily mean that nighttime detection has become easier. Differences in object frequencies, scene complexity, and visibility between the validation subsets may contribute to this result.

C. Effect of Training Data Composition

Balanced training also provides the strongest nighttime performance among the fine-tuned models. E1 exceeds E2 by 8.43 percentage points and E3 by 10.48 points in nighttime mAP@0.50.

E3 remains competitive during daytime, achieving 44.74% mAP@0.50 compared with E1's 45.23%. It also obtains a marginally higher daytime mAP@0.50:0.95 of 27.14%, compared with 27.04% for E1. This difference of 0.10 percentage points is too small to establish a meaningful advantage without repeated experiments.

These observations suggest that balanced training is associated with both stronger overall accuracy and a smaller difference between daytime and nighttime performance.

D. Limitations

Although the daytime and nighttime validation sets contain equal numbers of images, the distribution of object classes is not identical. Cars appear more frequently than bicycles, buses, and trucks, which may influence the overall detection results. Differences in object size, visibility, and scene complexity may also affect model performance.

Furthermore, the experiments focus on only five object categories from BDD100K, so the findings may not fully represent detection performance across all road objects. Finally, the reported results are based on a single validation run. Evaluating the models on the held-out test set and repeating the experiments would provide a more reliable assessment of their performance on unseen data.

V. CONCLUSION

This study investigated how daytime and nighttime training images affect the object detection performance of YOLOv8n on BDD100K. The results show that fine-tuning improves detection accuracy compared with the COCO-pretrained baseline, particularly for nighttime scenes.

Among the evaluated configurations, balanced training achieved the strongest overall performance, reaching 45.23% daytime and 47.91% nighttime mAP@0.50. In comparison, models trained exclusively on daytime or nighttime images

showed lower performance when evaluated under the opposite lighting condition. These findings suggest that including both lighting conditions during training can improve detection performance across different driving environments.

Future work could evaluate the models on the held-out test set, investigate additional object categories, and explore data augmentation techniques to improve detection under challenging lighting conditions.

REFERENCES

[1] F. Yu et al., "BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 2636–2645.

[2] Ultralytics, "YOLOv8 Documentation." [Online]. Available: https://docs.ultralytics.com/models/yolov8/. Accessed: Oct. 24, 2026.

[3] T.-Y. Lin et al., "Microsoft COCO: Common Objects in Context," in *Computer Vision–ECCV 2014*, Springer, 2014, pp. 740–755.

[4] Y. P. Loh and C. S. Chan, "Getting to Know Low-Light Images with The Exclusively Dark Dataset," *Computer Vision and Image Understanding*, vol. 178, pp. 30–42, 2019.